COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithm	**SECTION:**	CPE21S1
DATE PERFORMED:	07/09/2026	**DATE SUBMITTED:**	
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ACTIVITY NO. 1.1 **Basic C++ Programming**

1. PROCEDURE OUTPUT

- Create a function that will take two numbers and display the sum.

```cpp
#include <iostream>

int sumOfTwoNumbers(int fNum, int INum);

void sumTwoNum(int fNum, int INum);

int main() {

    std::cout << sumOfTwoNumbers(6, 7) << std::endl;

    sumTwoNum(7, 11);

    return 0;
}

int sumOfTwoNumbers(int fNum, int INum) {

    int total = fNum + INum;

    return total;
}

void sumTwoNum(int fNum, int INum) {

    int total = fNum + INum;

    std::cout << "Void Output : " << total << std::endl;
}
```

Sections	Answers
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`int sumOfTwoNumbers(int fNum, int INum);` `void sumTwoNum(int fNum, int INum);`
Class Declaration and Method Definition Section	NONE

Main Function	`int main() {`
	`  std::cout << sumOfTwoNumbers(6, 7) << std::endl;`
	`  sumTwoNum(7, 11);`
	`  return 0;`
	`}`

Method Definition	`int sumOfTwoNumbers(int fNum, int INum) {`
	`  int total = fNum + INum;`
	`  return total;`
	`}`
	`void sumTwoNum(int fNum, int INum) {`
	`  int total = fNum + INum;`
	`  std::cout << "Void Output : " << total << std::endl;`
	`}`

- Create a function that will return whether variable A is greater than variable B.

```
#include <iostream>

void greaterNum(int fNum, int INum);

int main() {

    int fNum, INum;

    std::cout<<"What Number is Greater Calculator"<<std::endl;
    std::cout<<"\nEnter First Number"<<std::endl;
    std::cin >> fNum;
```

```
    std::cout<<"\nEnter Second Number"<<std::endl;
    std::cin >> INum;

    greaterNum(fNum, INum);


    return 0;
}

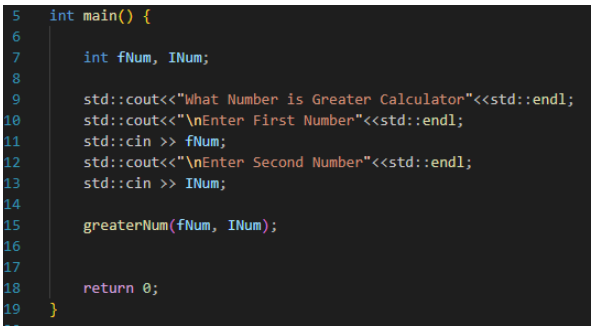
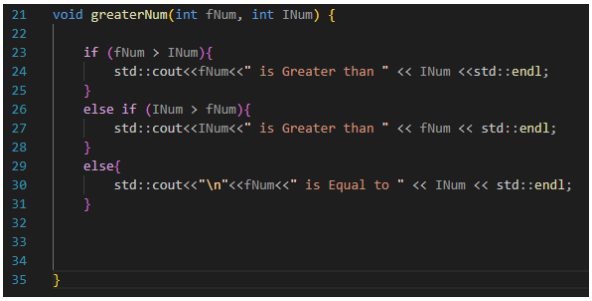
void greaterNum(int fNum, int INum) {

    if (fNum > INum){
        std::cout<<fNum<<" is Greater than " << INum <<std::endl;
    }
    else if (INum > fNum){
        std::cout<<INum<<" is Greater than " << fNum << std::endl;
    }
    else{
        std::cout<<"\n"<<fNum<<" is Equal to " << INum << std::endl;
    }



}
```

Sections	Answers
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`void greaterNum(int fNum, int INum);`
Class Declaration and Method Definition Section	NONE

Main Function	
Method Definition	

```
int main() {

    int fNum, INum;

    std::cout<<"What Number is Greater Calculator"<<std::endl;
    std::cout<<"\nEnter First Number"<<std::endl;
    std::cin >> fNum;
    std::cout<<"\nEnter Second Number"<<std::endl;
    std::cin >> INum;

    greaterNum(fNum, INum);


    return 0;
}
```

```
void greaterNum(int fNum, int INum) {

    if (fNum > INum){
        std::cout<<fNum<<" is Greater than " << INum <<std::endl;
    }
    else if (INum > fNum){
        std::cout<<INum<<" is Greater than " << fNum << std::endl;
    }
    else{
        std::cout<<"\n"<<fNum<<" is Equal to " << INum << std::endl;
    }


}
```

- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

```
#include <iostream>

bool logicalOperations(bool firstBool, bool secondBool);

int main() {

    bool firstBool, secondBool;

    std::cout << "Logical Operations Calculator" << std::endl;

    std::cout << "\nEnter First Boolean (0 for False, 1 for True): " <<
std::endl;
    std::cin >> firstBool;

    std::cout << "\nEnter Second Boolean (0 for False, 1 for True): " <<
std::endl;
    std::cin >> secondBool;
```

```
    if (logicalOperations(firstBool, secondBool)) {
        std::cout << "\nFunction executed successfully." << std::endl;
    }

    return 0;
}

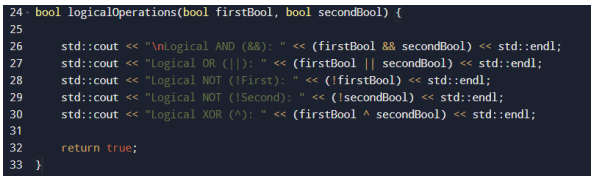
bool logicalOperations(bool firstBool, bool secondBool) {

    std::cout << "\nLogical AND (&&): " << (firstBool && secondBool) <<
std::endl;
    std::cout << "Logical OR (||): " << (firstBool || secondBool) <<
std::endl;
    std::cout << "Logical NOT (!First): " << (!firstBool) << std::endl;
    std::cout << "Logical NOT (!Second): " << (!secondBool) << std::endl;
    std::cout << "Logical XOR (^): " << (firstBool ^ secondBool) <<
std::endl;

    return true;
}
```

Sections	Answers
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	`bool logicalOperations(bool firstBool, bool secondBool);`
Class Declaration and Method Definition Section	NONE

Main Function	`bool firstBool, secondBool;`
	
	`std::cout << "Logical Operations Calculator" << std::endl;`
	
	`std::cout << "\nEnter First Boolean (0 for False, 1 for True): " << std::endl;`
	`std::cin >> firstBool;`
	
	`std::cout << "\nEnter Second Boolean (0 for False, 1 for True): " << std::endl;`
	`std::cin >> secondBool;`
	
	`if (logicalOperations(firstBool, secondBool)) {`
	`  std::cout << "\nFunction executed successfully." << std::endl;`
	`}`
	
	`return 0;`
	`}`

Method Definition	

```
bool logicalOperations(bool firstBool, bool secondBool) {

    std::cout << "\nLogical AND (&&): " << (firstBool && secondBool) << std::endl;
    std::cout << "Logical OR (||): " << (firstBool || secondBool) << std::endl;
    std::cout << "Logical NOT (!First): " << (!firstBool) << std::endl;
    std::cout << "Logical NOT (!Second): " << (!secondBool) << std::endl;
    std::cout << "Logical XOR (^): " << (firstBool ^ secondBool) << std::endl;

    return true;
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

The supplementary activities are meant to gauge your ability in using C++. The problems below range from easy to intermediate to advanced problems. Note your difficulties after answering the problems below.

1. Create a C++ program to swap the two numbers in different variables.

```
#include <iostream>

int main() {
    int a, b, temp;

    std::cout << "Enter first number: ";
    std::cin >> a;

    std::cout << "Enter second number: ";
    std::cin >> b;

    temp = a;
    a = b;
    b = temp;

    std::cout << "\nAfter Swapping:" << std::endl;
    std::cout << "First Number: " << a << std::endl;
    std::cout << "Second Number: " << b << std::endl;

    return 0;
}
```

- My difficulty was choosing between using pointers or the easier method. I decided to use a temporary

integer variable because I am more confident using it. However, I will continue studying pointers so I can improve my understanding and further enhance my coding skills.

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```
#include <iostream>
using namespace std;

double kelvinToFahrenheit(double kelvin)
{
    return (kelvin - 273.15) * 9 / 5 + 32;
}

double fahrenheitToKelvin(double fahrenheit)
{
    return (fahrenheit - 32) * 5 / 9 + 273.15;
}

int main()
{
    int choice;
    double temp;

    cout << "Temperature Converter" << endl;
    cout << "1. Kelvin to Fahrenheit" << endl;
    cout << "2. Fahrenheit to Kelvin" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice)
    {
        case 1:
            cout << "Enter temperature in Kelvin: ";
            cin >> temp;
            cout << "Fahrenheit: " << kelvinToFahrenheit(temp) << " F";
            break;

        case 2:
            cout << "Enter temperature in Fahrenheit: ";
            cin >> temp;
            cout << "Kelvin: " << fahrenheitToKelvin(temp) << " K";
```

```
            break;
#include <iostream>

double kelvinToFahrenheit(double kelvin)
{
    return (kelvin - 273.15) * 9 / 5 + 32;
}

double fahrenheitToKelvin(double fahrenheit)
{
    return (fahrenheit - 32) * 5 / 9 + 273.15;
}

int main()
{
    int choice;
    double temp;

    std::cout << "Temperature Converter\n";
    std::cout << "1. Kelvin to Fahrenheit\n";
    std::cout << "2. Fahrenheit to Kelvin\n";
    std::cout << "Enter your choice: ";
    std::cin >> choice;

    switch (choice)
    {
        case 1:
            std::cout << "Enter temperature in Kelvin: ";
            std::cin >> temp;
            std::cout << "Temperature in Fahrenheit: "
                      << kelvinToFahrenheit(temp) << " F\n";
            break;

        case 2:
            std::cout << "Enter temperature in Fahrenheit: ";
            std::cin >> temp;
            std::cout << "Temperature in Kelvin: "
                      << fahrenheitToKelvin(temp) << " K\n";
            break;
```

```
        default:
            std::cout << "Invalid choice!\n";
    }

    return 0;
}
```

- This one was pretty easy for me to do. The only thing that I forgot is the formula. I did this pretty confidently because I've had notes of doing this from my previous c++ class.

3. Create a C++ program that has a function that will calculate the distance between two points.

```
#include <iostream>
#include <cmath>

double calculateDistance(double x1, double y1, double x2, double y2);

int main() {
    double x1, y1, x2, y2;

    std::cout << "Enter x1: ";
    std::cin >> x1;

    std::cout << "Enter y1: ";
    std::cin >> y1;

    std::cout << "Enter x2: ";
    std::cin >> x2;

    std::cout << "Enter y2: ";
    std::cin >> y2;

    std::cout << "Distance = " << calculateDistance(x1, y1, x2, y2);

    return 0;
}

double calculateDistance(double x1, double y1, double x2, double y2) {
    return std::sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
}
```

- In this I really find it difficult since I'm bad at math but then again I researched the formula and it was kinda easy after that.

4. Modify the code given in ILO B and add the following functions:

a. A function to compute for the area of a triangle

```
#include <iostream>
#include <cmath>

double triangleArea(double base, double height);

int main() {
    double base, height;

    std::cout << "Enter the base of the triangle: ";
    std::cin >> base;

    std::cout << "Enter the height of the triangle: ";
    std::cin >> height;

    std::cout << "Area of the triangle: " << triangleArea(base, height);

    return 0;
}

double triangleArea(double base, double height) {
    return (base * height) / 2;
}
```

- It's average difficulty for me. The formula is the only thing that slows me down while creating this program.

b. A function to compute for the perimeter of a triangle

```
#include <iostream>

double trianglePerimeter(double side1, double side2, double side3);

int main() {
    double side1, side2, side3;

    std::cout << "Enter side 1: ";
```

```
    std::cin >> side1;

    std::cout << "Enter side 2: ";
    std::cin >> side2;

    std::cout << "Enter side 3: ";
    std::cin >> side3;

    std::cout << "Perimeter of the triangle: " << trianglePerimeter(side1,
side2, side3);

    return 0;
}

double trianglePerimeter(double side1, double side2, double side3) {
    return side1 + side2 + side3;
}
```

- This one is similar to the first one, it was easy to do but the formula makes it complicated, though this one's formula was pretty easy.

c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```
#include <iostream>

void triangleType(double a, double b, double c);

int main() {
    double a, b, c;

    std::cout << "Enter side 1: ";
    std::cin >> a;

    std::cout << "Enter side 2: ";
    std::cin >> b;

    std::cout << "Enter side 3: ";
    std::cin >> c;

    triangleType(a, b, c);
```

```
    return 0;
}

void triangleType(double a, double b, double c) {
    double temp;

    if (a > b) {
        temp = a;
        a = b;
        b = temp;
    }

    if (b > c) {
        temp = b;
        b = c;
        c = temp;
    }

    if (a > b) {
        temp = a;
        a = b;
        b = temp;
    }

    double left = (a * a) + (b * b);
    double right = c * c;

    if (left > right) {
        std::cout << "The triangle is Acute angled.";
    } else if (left < right) {
        std::cout << "The triangle is Obtuse angled.";
    } else {
        std::cout << "The triangle is Others.";
    }
}
```

- This one was very hard for me. It's because of the formula and sometimes it errors even though I did it well. This one took a lot of time in researching the formulas and how to fix the errors in the code.

FULL CODE:

```cpp
#include <iostream>
#include <cmath>

double triangleArea(double base, double height);
double trianglePerimeter(double side1, double side2, double
side3);
void triangleType(double side1, double side2, double side3);

int main() {
    int choice;
    double base, height;
    double side1, side2, side3;

    std::cout << "Triangle Calculator\n";
    std::cout << "1. Compute Area\n";
    std::cout << "2. Compute Perimeter\n";
    std::cout << "3. Determine Triangle Type\n";
    std::cout << "Enter your choice: ";
    std::cin >> choice;

    switch (choice) {
        case 1:
            std::cout << "\nEnter the base: ";
            std::cin >> base;

            std::cout << "Enter the height: ";
            std::cin >> height;

            std::cout << "Area of the triangle: "
                      << triangleArea(base, height);
            break;

        case 2:
            std::cout << "\nEnter side 1: ";
```

```
            std::cin >> side1;

            std::cout << "Enter side 2: ";
            std::cin >> side2;

            std::cout << "Enter side 3: ";
            std::cin >> side3;

            std::cout << "Perimeter of the triangle: "
                      << trianglePerimeter(side1, side2,
side3);
            break;

        case 3:
            std::cout << "\nEnter side 1: ";
            std::cin >> side1;

            std::cout << "Enter side 2: ";
            std::cin >> side2;

            std::cout << "Enter side 3: ";
            std::cin >> side3;

            triangleType(side1, side2, side3);
            break;

        default:
            std::cout << "Invalid choice.";
    }

    return 0;
}

double triangleArea(double base, double height) {
```

```
    return (base * height) / 2;
}

double triangle Perimeter(double side1, double side2, double
side3) {
    return side1 + side2 + side3;
}

void triangleType(double side1, double side2, double side3) {
    double a = side1;
    double b = side2;
    double c = side3;
    double temp;

    if (a > b) {
        temp = a;
        a = b;
        b = temp;
    }

    if (b > c) {
        temp = b;
        b = c;
        c = temp;
    }

    if (a > b) {
        temp = a;
        a = b;
        b = temp;
    }

    double left = (a * a) + (b * b);
    double right = c * c;
```

```
    if (left > right) {
        std::cout << "The triangle is Acute-angled.";
    } else if (left < right) {
        std::cout << "The triangle is Obtuse-angled.";
    } else {
        std::cout << "The triangle is Others (Right
triangle).";
    }
}
```

- So here I just combined everything using a switch case function, It was kinda easy and kinda hard but I think I did good.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGpt, QuilBot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **ChatGpt for formulas and checking my code errors and how to fix it, or what am I doing wrong. While I used quilBot for checking my grammar.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I still used my own thoughts creating the code but when I really don't know what to do anymore I ask ai what am I doing wrong. I didn't ask AI to make me a code.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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